BFGS on Convex Nonsmooth Functions: Local Convergence and Failure

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Abstract We study unmodified full BFGS on finite convex nonsmooth objectives, with Armijo–weak-Wolfe steps and actual gradients at differentiable accepted iterates. The main contrast concerns local geometry. A finite polyhedral partition near a unique minimizer, with locally Lipschitz gradient extensions on its cells, gives a common local convergence basin for every positive definite initial matrix. In contrast, a maximum of four globally convex $C^{2,1}$ functions with uniformly bounded Hessians admits deterministic nonstationary limits from starting points arbitrarily close to its unique minimizer. A separate construction retains uniform strong convexity with $C^{1,1}$ branches that lose second differentiability at the accumulation point. All accepted and rejected queries in these examples are differentiable and can remain arbitrarily close to the optimizer. The positive analysis uses a reference-energy identity, a hierarchy of actual principal minors of the full BFGS matrices, and reciprocal growth under bounded gradients and vanishing predicted decreases. A transported comparison certificate then relates this matrix budget to the objective's geometry. Neither the positive analysis nor the counterexample construction presumes iterate convergence or a matrix limit. Exact rational certificates, infinite scaling inductions, and a reviewed Lean correspondence support the proofs; the formal audit is distinguished from novelty and prose verification.

Keywords BFGS · convex nonsmooth optimization · local convergence · counterexample · formal verification

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1 Introduction and main results

Can ordinary BFGS converge locally on a convex objective whose gradients jump? Regularity of the formulas defining the objective does not by itself settle this question. We prove local convergence when a finite polyhedral arrangement controls the nearby smooth pieces, and construct local failure when four smooth convex formulas encode an accumulating sequence of bad switching configurations. The distinction is between the geometry of switching and the regularity within individual branches. It persists when all gradients queried by the algorithm are ordinary derivatives.

Throughout, BFGS means the unmodified full inverse update. An infinite *admissible run* starts with any positive definite matrix, uses fixed $0 < c_1 <$

$c_2 < 1$, satisfies Armijo decrease and weak Wolfe curvature, and accepts only differentiable points with nonzero gradients; the equations are given in Definition 1. This notion describes successful successive line searches. It does not guarantee that every search ray contains a differentiable acceptable point. The negative results instead verify every decision of one specified unit-first doubling/bisection implementation.

The following theorem gives the main positive contrast. The local extensions are defined on open sets, so smoothness at cell boundaries has a precise meaning.

Theorem 1 (Local convergence on finite polyhedral cells) *Let $n \geq 1$ and let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with a unique minimizer x_*. Suppose an open neighborhood U of x_* is covered by finitely many closed full-dimensional polyhedra P_i. For each i, suppose there is an open set $V_i \supset U \cap P_i$ and a function $f_i \in C^{1,1}_{\mathrm{loc}}(V_i)$ such that*

$$F(x) = f_i(x) \qquad (x \in U \cap P_i).$$

Then there exists $\varepsilon > 0$ such that every infinite full-BFGS run with arbitrary $H_0 \succ 0$, differentiable accepted iterates, fixed Armijo–weak-Wolfe constants $0 < c_1 < c_2 < 1$, and $F(x_0) \leq F(x_*) + \varepsilon$ satisfies

$$x_k \in U \quad (k \geq 0), \qquad F(x_k) \to F(x_*), \qquad x_k \to x_*.$$

The basin is chosen before the run and its initial matrix. Only the full objective must be convex; the extensions need not be convex outside their cells.

Finite maxima of smooth functions need not satisfy this finite-cell hypothesis. The next result fixes the objective and implementation before choosing a sequence of failing initial points. Its proof constructs all trial points, not only an accepted orbit.

Theorem 2 (Local failure with globally smooth convex branches) *There are four globally convex functions $f_i \in C^{2,1}(\mathbb{R}^2)$ such that*

$$0 \preceq \nabla^2 f_i(x) \preceq 64I \qquad (i = 0, 1, 2, 3; x \in \mathbb{R}^2)$$

and $F = \max_i f_i$ has compact sublevel sets and a unique nonsmooth minimizer at zero. For this one objective there are fixed starting points $x_0^{(n)} \to 0$, all with $H_0 = I$, such that, for every fixed pair

$$0 < c_1 \leq \tfrac{1}{3}, \qquad c_1 < c_2 < 1,$$

the deterministic unit-first bracketing implementation of Section 6 produces infinite full-BFGS runs satisfying

$$x_k^{(n)} \to p_n \neq 0, \qquad F(x_k^{(n)}) \downarrow F(p_n) > F(0) = 0, \qquad 0 \notin \partial F(p_n).$$

Every queried point is differentiable. Each line search takes at most five trials, and its accepted lengths repeat $1, 1/4, 1/16$. The objective, initial points, initial matrix and complete decision tree are the same throughout the displayed parameter range. The supremum of the norms of all accepted and rejected query points in the nth run tends to zero as $n \to \infty$.

Property	Theorem 2	Theorem 3, finite-branch version
Number of branches	Four	Four
Branch regularity	Global $C^{2,1}$	Global $C^{1,1}$
Positive lower curvature	Not imposed	Uniformly strongly convex
At the accumulation point	Branch Hessians exist	Three branch Hessians fail to exist
Initialization and locality	Fixed $H_0 = I$; all queries approach the optimizer	Fixed $H_0 = I$; all queries approach the optimizer

Table 1 Two local counterexamples along different regularity axes. The last row concerns a sequence of runs for one fixed objective in each column.

Theorem 2 rules out a local guarantee based only on a finite number of very smooth convex branch formulas. It does not say that every nearby starting point fails. The obstruction is compatible with compact sublevels and a unique minimizer. An independent construction changes the balance between smoothness and lower curvature.

Theorem 3 (Independent local failure with positive lower curvature) *There is a finite strongly convex function $F : \mathbb{R}^2 \to \mathbb{R}$ with compact sublevels and a unique nonsmooth minimizer at zero such that the following holds. For the fixed deterministic unit-first bracketing implementation of Section 6, with $c_1 = 10^{-4}$, $c_2 = 1/2$ and $H_0 = I$, there are starting points $x_0^{(n)} \to 0$ whose infinite full-BFGS runs converge to nonstationary points $p_n \neq 0$, with $p_n \to 0$ and $F(x_k^{(n)}) \downarrow F(p_n) > F(0)$. Every accepted and rejected query is differentiable, the accepted lengths repeat $1, 1/4, 1/16$, and the supremum of all query norms in the nth run tends to zero.*

The objective can be a maximum of four globally $C^{1,1}$, uniformly strongly convex branches with globally Lipschitz gradients. There are fixed $0 < \underline{\mu} \leq \overline{L} < \infty$ such that

$$\underline{\mu} I \preceq \nabla^2 F(x) \preceq \overline{L} I \quad \text{at every twice-differentiable point.}$$

The first three branches in this construction are not twice differentiable at zero. Alternatively, F can have one fixed positive-definite classical Hessian wherever that Hessian exists, with a convex countably piecewise-affine remainder. This alternative does not assert a finite smooth-branch representation.

In this last statement, a classical Hessian means the derivative of the gradient map, with that map defined on a neighborhood of the point. An isolated second Peano expansion is not the convention used. Table 1 keeps the incomparable regularity conclusions separate. In particular, neither construction gives local failure for a finite maximum of uniformly strongly convex C^2 branches through the optimizer. The common quadratic example used inside the second proof fails globally near a wrong corner; it is the subsequent embedding that gives failure near its own minimizer.

The positive proof starts from a comparison energy. Its negative term measures progress still visible to a supporting gradient; its positive term

measures a secant correction. We establish a hierarchy for the actual principal minors of full-BFGS matrices and derive divergence of the reciprocal arithmetic mean of $g_k^T H_k g_k$. This provides a budget that excludes a persistent objective gap when the objective supplies a suitable comparison field. The direction toward a fixed minimizer gives the finite-cell result. The criterion is sufficient, and is not claimed to characterize all convergent BFGS runs.

1.1 Related work and the role of the present results

Classical BFGS analysis combines convexity with matrix potentials and line-search inequalities. Powell's convergence analysis, the role of the quasi-Newton matrices studied by Byrd–Nocedal, and Nocedal's survey provide the relevant smooth background [11,10,9]. The Gram and Schur identities used below are standard matrix algebra. Our use of adjacent principal-minor degrees supplies a bridge from the full matrix update to a reciprocal budget in the bounded-gradient, vanishing-decrease regime. We do not attribute the underlying determinant, trace, or AM–GM tools to this work.

For unmodified nonsmooth BFGS, Guo–Lewis proves convergence on the Euclidean norm in every finite dimension [7, Corollary 4.2]; this is a known special case of the centered homogeneous setting below. Lewis–Overton studies the practical method and the line search used in our examples [1]. The finite two-piece affine escape theorem of (Xie and Wächter, 2017 preprint, arXiv:1712.08571v1, Theorem IV.1) is also prior work, with an unbounded-descent conclusion rather than the compact-level conclusion considered here. Gebken's criticality result assumes an iterate limit and specified matrix behavior [3]; those are not premises in our convergence theorems.

Earlier negative examples have different combinations of objective geometry and algorithmic hypotheses. Smooth nonconvex BFGS failures [17,18], the examples of (Lewis and Overton, *Behavior of BFGS with an Exact Line Search on Nonsmooth Examples*, 2008 manuscript), and other nonconvex quasi-Newton constructions [15] are relevant antecedents, but do not establish the two convex local statements above. Limited-memory failure results [16] concern a different update. The recent smooth nonconvex planar example of (Diao, 2026 preprint, arXiv:2609.09686v1) likewise has different objective assumptions. We compare these statements by their hypotheses; no claim of failure for every weak-Wolfe implementation is made.

The geometric interpretation of surviving smooth directions is related to VU theory, partial smoothness, and stratification [14,5,4]. Convex-gradient cocoercivity is an established ingredient [2]. The companion develops affine-critical and fixed-normal extensions using those antecedents. Such connections identify standard tools and related structures, without supplying a necessary-and-sufficient dynamical classification or establishing priority for a complete theorem.

Sections 2–4 develop the geometric motivation and the positive proof. Section 5 identifies the separate companion results and their limits. Section 6

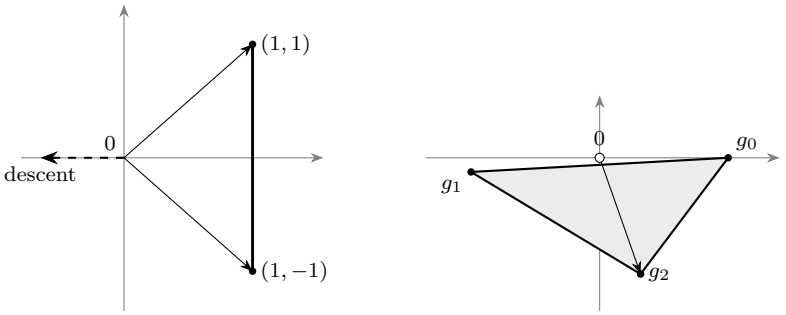
proves Theorem 2, with its finite certificate in Appendix B; Appendix C proves Theorem 3. All essential arguments for the article's theorems are contained here, apart from the explicitly invoked standard background theorems.

2 Geometry of progress and stagnation

The mixed ridge $F(u,v) = u^2 + |v|$ illustrates why a gradient jump need not remove useful descent. Along $v = 0$ its limiting gradients are $(2u, 1)$ and $(2u, -1)$. Their difference is vertical. At a point with $u \neq 0$, the horizontal vector $(-u, 0)$ has the same strictly negative pairing $-2u^2$ with both gradients. The inward radial model also has a small error:

$$F((1-\tau)(u,v)) - F(u,v) + \tau \nabla F(u,v)^T (u,v) = \tau^2 u^2 \qquad (v \neq 0,\ 0 \leq \tau \leq 1).$$

Thus the nonsmooth part is exactly linear under inward scaling. This is the mechanism formalized by the radial theorem, not an assertion about the eventual eigenvectors or eigenvalues of a computed BFGS matrix.



(a) $u^2 + |v|$, $u = 1/2$ (b) Certified nonstationary seed

Fig. 1 Gradient space. Left: the mixed ridge at $u = 1/2$ has a vertical jump and a common horizontal descent direction. Right: the particular three-gradient seed used in Section 6. Its convex hull excludes zero, but its affine hull is the plane. The vertices on the right are rendered from the certified root's rational center; the mathematical definition uses the isolated root. Arrows indicate gradients, not BFGS steps.

The bad corner in Figure 1 is specific. Its slopes are $g_0 = (1, 0)$, $g_1 = (-1, -1/8)$ and $g_2 = (c, d)$, where (c, d) is defined exactly in Appendix B. They are affinely independent, so no nonzero direction pairs equally with all three. Yet $(-1, 16)$ has negative pairing with each, and zero is outside their convex hull. The corner is therefore nonstationary. Section 6 shows that the actual algorithm nevertheless contracts through a three-step cycle toward this corner, with all line-search comparisons strict.

This distinction between affine and convex balance helps interpret the examples. For a nonempty finite family of slopes a_i, equality $a_i^T v = \kappa < 0$ for every i asks for a common rate of descent, whereas $a_i^T v < 0$ asks only for

descent. The former fails when zero belongs to their affine hull; the latter fails when zero belongs to their convex hull. Indeed, a linear functional constant on the slopes is constant on their affine hull. Conversely, if zero is outside that affine hull, its nearest point gives a separating functional constant on the hull; changing its sign gives the negative common value. The convex version follows from the same closest-point argument for the convex hull. These are elementary separation facts, not a classification of BFGS behavior.

At the minimizer of the mixed ridge there is no remaining objective gap to eliminate. At the certified wrong corner, contraction occurs while a strict descent direction remains; its coercive realization in Corollary 3 also has a positive limiting objective gap. A finite number of smooth formulas can repeat the latter configuration at infinitely many shrinking scales. The local positive result instead controls that repetition by a finite polyhedral partition near the optimizer. The proof makes this control quantitative through the error in an inward first-order model.

3 A common BFGS energy principle

The useful comparison quantity measures displacement in the current inverse metric. Its change separates progress toward a reference point from the cost of the new secant. We first derive this identity, and then show that the BFGS update forces enough reciprocal growth to rule out a persistent objective gap when the secant cost is controlled.

Definition 1 (Admissible runs) Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with $n \geq 1$. An admissible run is an infinite sequence of differentiable accepted points x_k, nonzero gradients $g_k = \nabla F(x_k)$, and matrices generated by

$$\begin{gathered} s_k = -\alpha_k H_k g_k, \qquad x_{k+1} = x_k + s_k, \qquad y_k = g_{k+1} - g_k, \\ H_{k+1} = \left(I - \frac{s_k y_k^T}{t_k}\right) H_k \left(I - \frac{y_k s_k^T}{t_k}\right) + \frac{s_k s_k^T}{t_k}, \qquad t_k = y_k^T s_k, \end{gathered} \tag{1}$$

where $H_0 \succ 0$, $\alpha_k > 0$, and fixed constants $0 < c_1 < c_2 < 1$ satisfy

$$F(x_{k+1}) \leq F(x_k) + c_1 g_k^T s_k, \qquad g_{k+1}^T s_k \geq c_2 g_k^T s_k. \tag{2}$$

These are the usual actual-gradient, full-BFGS conventions [1]; see also (Xie and Wächter, 2017 preprint, arXiv:1712.08571v1). A zero gradient is successful termination. Statements about admissible runs do not assert that a differentiable acceptable point exists on every ray, or that every implementation finds one.

Write

$$\begin{gathered} B_k = H_k^{-1}, \qquad q_k = g_k^T H_k g_k, \qquad d_k = -g_k^T s_k = \alpha_k q_k, \\ \Delta_k = F(x_k) - F(x_{k+1}), \qquad c = 1 - c_2. \end{gathered}$$

Lemma 1 *Every admissible run satisfies* $H_k \succ 0$, $q_k, d_k, t_k > 0$ *and*

$$c_1 d_k \le \Delta_k \le d_k, \qquad t_k \ge c d_k. \tag{3}$$

Moreover,

$$B_{k+1} = B_k - \frac{g_k g_k^T}{q_k} + \frac{y_k y_k^T}{t_k}, \qquad M_k := B_k - \frac{g_k g_k^T}{q_k} \succeq 0, \qquad M_k s_k = 0. \tag{4}$$

If the objective values have a finite limit, then $\sum_k d_k < \infty$.

Proof Given $H_k \succ 0$, nonzero g_k gives $q_k, d_k > 0$; weak Wolfe gives $t_k \ge (1 - c_2) d_k > 0$. For any vector v, the quadratic form in (1) is a sum of two nonnegative terms. If both vanish, then $s_k^T v = 0$ and $(I - y_k s_k^T / t_k) v = 0$, whence $v = 0$. Thus $H_{k+1} \succ 0$, proving the induction. Armijo gives the first lower bound in (3); the supporting inequality for F gives the upper bound. Summing the lower bound proves summability when the values have a finite limit.

The Hessian identity is the classical equivalent form of BFGS [9, Section 5]; it also follows by multiplying its right side by (1), using $B_k s_k = -\alpha_k g_k$ and $s_k^T y_k = t_k$. Finally, metric Cauchy–Schwarz gives $(g_k^T v)^2 \le q_k v^T B_k v$, and $g_k^T s_k = -\alpha_k q_k$ gives $M_k s_k = 0$.

Lemma 2 (Reference energy) *For any fixed* x_*, *put* $r_k = x_k - x_*$ *and*

$$E_k = r_k^T B_k r_k, \qquad a_k = g_k^T r_k, \qquad R_k = \frac{(y_k^T r_{k+1})^2}{t_k}.$$

Then

$$E_{k+1} = E_k - \frac{a_k^2}{q_k} + R_k. \tag{5}$$

If x_* *minimizes* F, *with* $F_* = F(x_*)$, *then*

$$(F(x_N) - F_*)^2 \sum_{k<N} q_k^{-1} \le E_0 + \sum_{k<N} R_k. \tag{6}$$

Proof Since $r_{k+1} = r_k + s_k$ and $M_k s_k = 0$, the Hessian update gives

$$r_{k+1}^T B_{k+1} r_{k+1} = r_k^T M_k r_k + \frac{(y_k^T r_{k+1})^2}{t_k},$$

which is (5). At a minimizer, convexity gives $a_k \ge F(x_k) - F_*$; the values decrease. Summation and $E_N \ge 0$ yield (6).

The negative term is progress measured through a supporting gradient. The positive term records the secant's component in the next reference direction. An upper bound on this correction becomes useful only when the reciprocal sum is known to grow sufficiently fast. That growth is forced by the full matrix update, without a spectral bound on the matrices.

3.1 Actual principal minors and reciprocal growth

For a symmetric $n \times n$ matrix H, define

$$e_j(H) = \sum_{\substack{I \subset \{1,\dots,n\} \\ |I|=j}} \det H_{I,I}, \qquad e_{n+1}(H) = 0.$$

These are actual principal determinants. Multilinearity of the determinant identifies them with the coefficients of $\det(I + zH)$; hence they are invariant under orthogonal changes of coordinates and equal the elementary symmetric polynomials of the eigenvalues. In particular $e_1 = \mathrm{tr}$ and $e_n = \det$. The determinant endpoint is classical; trace/determinant potentials have long been used in BFGS analysis [9,10]. We need all adjacent degrees.

Theorem 4 (Principal-minor hierarchy) *Let $H \succ 0$, $s = -\alpha Hg$, $\alpha > 0$, and suppose*

$$q = g^T Hg > 0, \qquad \beta = -\frac{y^T Hg}{q} > 0.$$

For the inverse-BFGS update H_+ in (1)*,*

$$e_j(H_+) \le \frac{\|y\|^2}{\beta^2 q} e_{j+1}(H) + \frac{\alpha}{\beta} e_j(H), \qquad 1 \le j \le n. \tag{7}$$

At $j = n$ the exact identity is $\det H_+ = (\alpha/\beta) \det H$.

Proof Let Z have j orthonormal columns. The complete compression calculation in Lemma 6 gives

$$\det(Z^T H_+ Z) \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q} + \frac{\alpha}{\beta} \det(Z^T H Z),$$

where $\mathrm{Gram}_H(y, Z) = [y\ Z]^T H [y\ Z]$. Choose an orthonormal eigenbasis of H, with eigenvalues $\lambda_i > 0$, and write y_i for the coordinates of y in this basis. Summing over its j-column selections Z_I gives

$$\begin{aligned} \sum_{|I|=j} \det \mathrm{Gram}_H(y, Z_I) &= \sum_{|I|=j} \left(\prod_{i \in I} \lambda_i \right) \sum_{\ell \notin I} \lambda_\ell y_\ell^2 \\ &= \sum_{|J|=j+1} \left(\prod_{i \in J} \lambda_i \right) \sum_{\ell \in J} y_\ell^2 \le \|y\|^2 e_{j+1}(H). \end{aligned}$$

The first equality is the Schur complement of the diagonal compression; the second groups terms by $J = I \cup \{\ell\}$. Orthogonal invariance now proves (7). The last assertion follows from the full-size case in Lemma 6.

Theorem 5 (Reciprocal growth) *If an admissible run has bounded gradients and $d_k \to 0$, then*

$$\left(\prod_{k<N} q_k^{-1}\right)^{1/N} \longrightarrow +\infty, \qquad \frac{1}{N}\sum_{k<N} q_k^{-1} \longrightarrow +\infty. \tag{8}$$

Proof Choose $G > 0$ with $\|g_k\| \le G$, and put $K_0 = 4G^2/c^2$. For the actual run,

$$\beta_k = \frac{t_k}{d_k} \ge c, \qquad \|y_k\| \le 2G, \qquad \alpha_k = \frac{d_k}{q_k}.$$

Theorem 4 therefore yields the normalized hierarchy

$$q_k e_j(H_{k+1}) \le K_0 e_{j+1}(H_k) + \frac{d_k}{c} e_j(H_k). \tag{9}$$

This recurrence is derived for the run's principal determinants.

Fix $\eta > 0$ and choose J so that $d_k/c \le \eta$ for $k \ge J$. For $H \succ 0$ define

$$V_\eta(H) = \sum_{j=1}^{n} (K_0/\eta)^{j-1} e_j(H) > 0.$$

Multiply (9) by these weights and sum. The e_{j+1} terms contribute at most $\eta V_\eta(H_k)$, as do the remaining terms. Also $q_k \le G^2 \operatorname{tr} H_k \le G^2 V_\eta(H_k)$. Consequently

$$V_\eta(H_{k+1}) \le \frac{2\eta}{q_k} V_\eta(H_k), \qquad \prod_{k=J}^{J+m-1} q_k^{-1} \ge \frac{1}{G^2 V_\eta(H_J)(2\eta)^{m-1}}.$$

For the last bound, iterate through $m-1$ updates and apply the bound on q_{J+m-1} at the remaining endpoint. Taking mth roots gives a lower limit of at least $(2\eta)^{-1}$. A fixed prefix contributes a factor tending to one. Since η is arbitrary, the geometric mean diverges; AM–GM gives the arithmetic-mean conclusion.

Product estimates and AM–GM are established tools in BFGS analysis [8, Theorem 3.7][12, Appendices C–D]. Here the normalized hierarchy supplies them in the bounded-gradient, vanishing-decrease setting. Neither limit in (8) asserts that $q_k \to 0$ pointwise or that H_k converges.

3.2 Moving comparison directions

A reference direction can be allowed to move, provided that it continues to see descent, controls the incoming secant, and does not increase its energy too rapidly. The identity $M_k s_k = 0$ shows that movement along the executed step has no cost in the part of the update that remains after the negative rank-one term. A positive-semidefinite comparison field expresses these requirements without a choice of frame.

For differentiable x, z, write $g = \nabla F(x)$, $h = \nabla F(z)$, $s = z - x$, $y = h - g$, $d = -g^T s$, $\Delta = F(x) - F(z)$ and $t = y^T s$. Call the pair eligible when

$$d > 0, \qquad c_1 d \le \Delta \le d, \qquad t \ge cd.$$

All accepted pairs are eligible. Requiring a property of every eligible pair is an objective assumption; it does not assert that an arbitrary sequence of eligible pairs can be realized by BFGS. Put $\Pi_s = I - ss^T/(s^T s)$.

Definition 2 (Band certificate) Let $K = \{F \le b\}$ be compact, with $b > F_* = \min F$. For each nonempty $K_\eta = \{x \in K : F(x) \ge F_* + \eta\}$, $\eta > 0$, a band certificate consists of a finite-valued symmetric PSD field $W_\eta(x)$ on its differentiable points and constants $m_\eta > 0$, $C_\eta \ge 0$, $L_\eta \ge 0$, $\delta_\eta > 0$ such that

$$g^T W_\eta(x) g \ge m_\eta, \tag{10}$$

$$y^T W_\eta(z) y \le C_\eta t, \tag{11}$$

$$\Pi_s W_\eta(z) \Pi_s \preceq (1 + L_\eta d) \Pi_s W_\eta(x) \Pi_s. \tag{12}$$

The first inequality holds at every differentiable point of K_η; the last two hold for all eligible pairs in K_η with $d \le \delta_\eta$.

Neither continuity, uniform boundedness nor positive definiteness of W_η is required. The constants and field may depend on the positive band. Allowing an arbitrary real secant constant gives the same condition, since it can always be replaced by its maximum with zero.

Theorem 6 (Transported comparison certificate) *If K admits the band certificates above, every admissible run with $x_0 \in K$ satisfies*

$$F(x_k) \to F_*, \qquad \operatorname{dist}(x_k, \arg\min F) \to 0.$$

If the minimizer is unique, the iterates converge to it.

Proof Armijo keeps the run in K, and its values have a finite limit. Lemma 1 gives $\sum d_k < \infty$. Gradients are bounded there: choose $\epsilon > 0$ and bound the continuous finite convex function on $K + \epsilon\overline{B}$. Its supporting inequality at x evaluated at $x + \epsilon u$, for every unit u, bounds every subgradient at x. Thus Theorem 5 applies.

Suppose the limiting gap is $\ell > 0$. On a tail $k \ge J$, use the field on $K_{\ell/2}$ and ensure $d_k \le \delta_{\ell/2}$; omit the band subscripts. Set $E_k = \operatorname{tr}(B_k W(x_k)) \ge 0$. Since $M_k = \Pi_{s_k} M_k \Pi_{s_k} \succeq 0$, trace monotonicity and (11)–(12) give

$$\begin{aligned} E_{k+1} &= \operatorname{tr}(M_k W(x_{k+1})) + \frac{y_k^T W(x_{k+1}) y_k}{t_k} \\ &\le (1 + L d_k) \operatorname{tr}(M_k W(x_k)) + C \le (1 + L d_k)(E_k - m/q_k) + C. \end{aligned}$$

Trace monotonicity here follows from $\operatorname{tr}(AQ) = \operatorname{tr}(Q^{1/2}AQ^{1/2}) \geq 0$ for PSD A, Q. Let $P_J = 1$ and $P_{k+1} = P_k(1 + Ld_k)$. Summability gives $1 \leq P_k \leq \exp(L\sum_{j\geq J} d_j) =: \overline{P}$. Dividing by P_{k+1}, summing and using $E_N \geq 0$ yields

$$\frac{m}{\overline{P}} \sum_{k=J}^{N-1} q_k^{-1} \leq E_J + C(N-J),$$

contradicting (8). This normalization is the usual summable multiplicative-error argument; compare [13, Lemmas 2.2–2.3]. Finally, a sequence in compact K with values tending to F_* cannot stay a positive distance from the closed solution set: a convergent subsequence would have a nonoptimal limit with value F_*. With a unique minimizer the same argument gives convergence of the entire sequence.

4 Geometries guaranteeing local convergence

The radial direction supplies the simplest comparison field. It is especially natural for a homogeneous term: if γ is finite, convex and positively homogeneous of degree one, then

$$\gamma((1-\tau)r) = (1-\tau)\gamma(r), \qquad 0 \leq \tau < 1.$$

Its inward model is exact even when its angular derivatives jump. Adding a smooth term leaves a quadratic inward modeling error. We make this observation precise, prove its equivalence with radial secant control, and then use finite polyhedral geometry to obtain the local theorem stated in the introduction.

4.1 Inward models and radial secants

Fix a minimizer x_*, write $F_* = F(x_*)$, and let $K = \{x : F(x) \leq b\}$ be compact, with $b > F_*$. For differentiable $x, z \in K$, set

$$r_x = x - x_*, \qquad g = \nabla F(x), \qquad h = \nabla F(z),$$
$$T = (h-g)^T(z-x), \qquad N = (h-g)^T r_z.$$

The radial secant condition is the following requirement on positive bands:

$$\begin{gathered}\text{for every } \eta > 0 \text{ there is } C_\eta \geq 0 \text{ such that} \\ N^2 \leq C_\eta T \quad \text{whenever } F_* + \eta \leq F(z) \leq F(x), \quad T > 0.\end{gathered} \tag{13}$$

Here all such differentiable endpoint pairs are included, whether or not they are eligible for the fixed line search. The equivalent model condition is

$$\begin{gathered}\text{for every } \eta > 0 \text{ there are } M_\eta \geq 0,\ \theta_\eta \in (0, 1/2] \text{ such that} \\ 0 \leq F(x - \tau r_x) - F(x) + \tau \nabla F(x)^T r_x \leq \frac{M_\eta}{2}\tau^2, \quad 0 \leq \tau \leq \theta_\eta,\end{gathered} \tag{14}$$

at every differentiable $x \in K$ with $F(x) \geq F_* + \eta$. The inward point need not be differentiable. It remains in K by convexity. The parameter τ is a fraction of the distance to x_*, rather than a Euclidean step length.

Theorem 7 (Radial equivalence) *For finite convex F and the compact sublevel K above,* (13) *and* (14) *are equivalent.*

Proof Choose $G, R > 0$ bounding subgradient norms on K and $\|x - x_*\|$ there. The subgradient bound was proved in Theorem 6; supporting inequalities then give $|F(x) - F(z)| \leq G\|x - z\|$ on K. Use the supporting-plane error

$$D_F(z, x) = F(z) - F(x) - \nabla F(x)^T (z - x) \geq 0.$$

This is the usual Bregman expression [6]; only its supporting endpoint needs to be differentiable. At differentiable endpoints, $T = D_F(z, x) + D_F(x, z)$.

Suppose first that (14) holds, and fix a positive band, with constants M, θ. If $N \geq 0$, compare the supporting plane at x with the inward model at $z - \tau r_z$ to obtain

$$\tau N \leq D_F(z, x) + \tfrac{1}{2} M \tau^2.$$

If $N < 0$, reverse the endpoints and use the inward model at x:

$$\tau(T - N) \leq D_F(x, z) + \tfrac{1}{2} M \tau^2.$$

Thus there are $A \in [|N|, 2GR]$ and $D \in \{D_F(z, x), D_F(x, z)\}$ such that $\tau A \leq D + M\tau^2/2$ for $0 \leq \tau \leq \theta$. For $M > 0$ and $A \leq M\theta$, choose $\tau = A/M$, giving $A^2 \leq 2MD$. If $A > M\theta$, take $\tau = \theta$ to obtain $D \geq A\theta/2$ and hence $A^2 \leq (4GR/\theta)D$. When $M = 0$, the same endpoint choice gives an even stronger estimate. Therefore

$$N^2 \leq \max\{2M, 4GR/\theta\}\, T. \tag{15}$$

Conversely, assume (13), fix $\eta > 0$, and put

$$\theta = \min\{1/2, \eta/(4GR)\}.$$

For x on the η-band and $z = x - \tau r_x$, $0 < \tau \leq \theta$, convexity gives $F(z) \leq F(x)$ and the Lipschitz estimate gives $F(z) \geq F_* + 3\eta/4$. Suppose initially that z is differentiable and set $a = (\nabla F(x) - \nabla F(z))^T r_x \geq 0$. Then

$$T = \tau a, \qquad N = -(1 - \tau)a.$$

If $a > 0$, the secant bound on the $\eta/2$-band implies $a \leq C_{\eta/2}\tau/(1 - \tau)^2$. If $a = 0$, no division is needed. In both cases convexity yields

$$0 \leq D_F(z, x) \leq \tau a \leq 4C_{\eta/2}\tau^2.$$

To remove differentiability at z, use the standard fact that a finite convex function is differentiable almost everywhere [20, Theorem 25.5]. For fixed $0 < \tau < 1$, the dilation $x \mapsto x - \tau(x - x_*)$ is invertible; therefore points at which both the original and dilated points are differentiable form a full-measure,

hence dense, set. Approximate x by such points x_j in the interior of K. This interior is nonempty since $b > F_*$, and its closure is K: moving any point slightly toward x_* gives value strictly below b. The strict margin $3\eta/4 > \eta/2$ ensures that the approximating pairs lie on the required band for large j.

At a differentiability point x of a finite convex function, $\nabla F(x_j) \to \nabla F(x)$. Indeed, local subgradient bounds give compactness, and every subsequential gradient limit satisfies the supporting inequality at x, whose subgradient is unique. Passing to the limit in the preceding function-value inequality now proves (14) with $M_\eta = 8C_{\eta/2}$. The case $\tau = 0$ is immediate.

Corollary 1 (Radial convergence) *Under either equivalent condition, every admissible run starting in K converges in value to F_* and in distance to* $\arg\min F$. *With a unique minimizer it converges to x_*.*

Proof Take $W_\eta(x) = r_x r_x^T$. Convexity gives $g^T r_x \geq F(x) - F_* \geq \eta$, proving visibility with $m_\eta = \eta^2$. Equation (13) gives secant control on eligible pairs. Finally $r_z = r_x + s$ implies $\Pi_s r_z = \Pi_s r_x$, so transport holds with $L_\eta = 0$. Apply Theorem 6.

This condition controls only inward first-order modeling error. In a smooth region its infinitesimal expression is $r_x^T \nabla^2 F(x) r_x$, rather than the largest Hessian eigenvalue. A bound almost everywhere on that expression alone would miss a jump of the derivative along a ray; the function-value condition (14) includes such jumps.

4.2 Centered homogeneous terms and smooth perturbations

Theorem 8 (Centered smooth-plus-sublinear objectives) *Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, and suppose x_* minimizes F. Let U be an open convex set, let ψ be convex and locally $C^{1,1}$ on U, and let $\gamma : \mathbb{R}^n \to \mathbb{R}$ be finite, convex and positively homogeneous of degree one. Suppose*

$$F(x) = \psi(x) + \gamma(x - x_*) \qquad (x \in U).$$

Every admissible run whose initial sublevel is compact and contained in U satisfies $F(x_k) \to F(x_*)$. If x_* is the unique minimizer, then $x_k \to x_*$. Neither original summand is required to be minimized at x_*.

Proof Let $K = \{F \leq F(x_0)\}$. Since $g_0 \neq 0$, convexity implies $F(x_0) > F(x_*)$. On compact convex $K \subset U$, local $C^{1,1}$ regularity provides a uniform Lipschitz constant L for $\nabla\psi$: take a finite cover by neighborhoods carrying local constants and subdivide any segment in K into pieces in that cover. Summing the local bounds gives the maximum of those constants along the segment. Integrating the gradient along an inward segment consequently gives

$$\left|\psi(x - \tau r_x) - \psi(x) + \tau \nabla\psi(x)^T r_x\right| \leq \tfrac{1}{2} L \tau^2 \|r_x\|^2.$$

At a differentiable point of F in U, the representation and smoothness of ψ imply differentiability of γ at r_x. Differentiating its homogeneity identity gives $\nabla\gamma(r_x)^T r_x = \gamma(r_x)$. Its contribution to the inward remainder is therefore exactly zero. The upper bound in (14) follows with $M = L \max_{x\in K}\|r_x\|^2$; its lower bound follows from convexity of F. Corollary 1 proves the claim.

For $F(u, v) = u^2 + |v|$, take $\psi(u, v) = u^2$ and $\gamma(u, v) = |v|$. Every sublevel is compact and the minimizer is unique, so every admissible run converges to $(0, 0)$. Guo–Lewis proves the weaker subsequential conclusion for this example [7, Corollary 4.4]. Their Euclidean-norm convergence theorem [7, Corollary 4.2] is an established special case also recovered here. Lewis–Overton [1, Section 5.1] analyzes the scalar absolute-value model with its stated line-search conventions. The proof above uses inward modeling directly. Convex-gradient cocoercivity [2] supplies an alternative secant estimate, but no global Lipschitz-gradient hypothesis on F is required.

4.3 Local finite partitions

The distinction between a local objective hypothesis and a hypothesis on the whole initial sublevel is handled by a simple convex trapping argument.

Lemma 3 (A common local basin) *Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex with a unique minimizer x_*. Every sublevel is compact. For each neighborhood U of x_*, there is $\varepsilon > 0$ such that $\{F \le F(x_*) + \varepsilon\} \subset U$. Thus any admissible run starting in that sublevel stays in U, independently of its initial SPD matrix.*

Proof Choose $r > 0$ with $\overline{B}_r(x_*) \subset U$. Continuity, compactness of the sphere and uniqueness give

$$\omega = \min_{\|u\|=r} \big(F(x_* + u) - F(x_*)\big) > 0.$$

For $\|x - x_*\| \ge r$, convexity at the point at distance r on the segment from x_* to x gives

$$F(x) - F(x_*) \ge \frac{\|x - x_*\|}{r}\omega.$$

Consequently F is coercive, hence has compact sublevels, and $\varepsilon = \omega/2$ gives the asserted inclusion. Armijo preserves it.

Proof (Proof of Theorem 1) First discard cells not containing x_*. Each such closed cell has an open complement containing x_*; finiteness supplies one ball that avoids them all. For every remaining cell, its extension domain V_i contains x_*. Shrink to a common ball $B_r(x_*) \subset U \cap \bigcap_{i:x_*\in P_i} V_i$, and then shrink again so that the gradients of all remaining extensions have one Lipschitz bound L there. This uses only finitely many local extensions; they need not have shared a domain before shrinking.

Let x be a differentiability point of F in this ball, and choose a cell P_i containing it. Convexity of P_i places the whole inward segment $x - \tau(x - x_*)$,

$0 \le \tau \le 1$, in P_i. Both F and f_i agree along that segment. Comparing their right derivatives at $\tau = 0$ gives

$$\nabla F(x)^T(x - x_*) = \nabla f_i(x)^T(x - x_*).$$

Only equality in the radial direction is needed, including at cell boundaries. The smooth Taylor estimate along the same segment gives

$$0 \le F(x - \tau r_x) - F(x) + \tau \nabla F(x)^T r_x \le \tfrac{1}{2} L\tau^2 \|r_x\|^2 \le \tfrac{1}{2} L r^2 \tau^2.$$

The first inequality uses convexity of F, not convexity of the extension. Lemma 3 supplies a compact positive sublevel wholly contained in this ball, chosen before the run or its matrix. The inward condition holds on that sublevel, so Corollary 1 gives value and iterate convergence. Armijo gives the asserted containment.

Corollary 2 *A finite convex piecewise linear-quadratic objective with a unique minimizer has the local convergence property of Theorem 1. The same holds for a convex locally $C^{1,1}$ loss plus a finite convex piecewise linear-quadratic regularizer, provided the sum has a unique minimizer.*

Proof By piecewise linear-quadratic we mean a representation by quadratic polynomials on a finite cover by closed full-dimensional polyhedral cells. Each polynomial is a smooth extension of its cell formula; adding the smooth loss preserves the required local $C^{1,1}$ extensions. Theorem 1 applies.

This covers finite absolute-value and hinge penalties with locally $C^{1,1}$ convex losses, under the stated uniqueness assumption. Different cells may have different quadratic Hessians. A finite maximum of arbitrary quadratic formulas need not admit such a polyhedral partition and is not automatically covered. There is also no claim that iterates identify a cell or that rejected trials remain in the local structural neighborhood.

For a further nonpolyhedral example, consider a locally $C^{1,1}$ convex loss plus finitely many Euclidean residual norms $\|A_j x - b_j\|$. Near a minimizer, a term with nonzero residual is smooth, whereas a zero residual becomes $\|A_j(x - x_*)\|$, a centered sublinear term. Their sum has the local representation in Theorem 8. Under uniqueness, Lemma 3 supplies the common low-starting basin. The smoothness assertion here uses the Euclidean norm away from zero; it does not apply to every spectral penalty at a nonzero rank-deficient point.

5 Beyond radial geometry: the companion's scope

The radial criterion is a useful sufficient condition. It is not necessary for convergence, and the broader transported certificate is not necessary either. The complete companion report retains a scalar convergence baseline and a scalar certificate-nonexistence example that separate these questions. It also develops three distinct geometric extensions. These additional results are recorded here for orientation; they are not among the theorems independently established by this article.

The planar result in the companion's Chapter 9 concerns regular switching and strict positive-level contours. Its comparison direction moves along the contour geometry; the proof uses a signed determinant estimate and one-sided derivative comparison. Its nondegenerate smooth-max application does not require the radial secant condition. Chapter 11 treats fixed jump normals in arbitrary dimension, where a smooth-subspace residual budget replaces the planar argument. Chapter 10 treats a parabolic family with flat contour pieces; that case does not inherit the strict-contour hypothesis. The corresponding exact formal contracts and proofs are separate entries in the companion correspondence index.

These distinctions also limit the geometric interpretation. A certificate can annihilate accessible limiting gradient jumps under an extension hypothesis; it need not annihilate every active-gradient difference without an additional accessibility premise. Affine-criticality describes a geometric obstruction, while a particular failing run requires its own dynamical construction. The article uses neither a universal identification of active and accessible jumps nor an assumed convergence of its metric.

For readers needing quantitative or noncompact conclusions, the companion also retains the finite-horizon reciprocal bounds, superquadratic reciprocal sums under summable predicted decreases, noncompact homogeneous alternatives, affine-fiber lifting, and conservative rates. Their hypotheses and conclusions are not all ordered by implication. They remain separate from the compact-level qualitative route used for the article's local positive theorem.

6 Deterministic local failure despite smooth convex branches

We now construct one objective with bad starting points at every sufficiently small scale. The construction first realizes a contracting cycle on a maximum of three affine functions. Carefully chosen scalar functions then place copies of this same cycle near the unique minimizer of one fixed objective. Every query of the implemented line search, including rejected queries, remains in the corresponding affine region.

6.1 The implementation and the local statement

At a differentiable point x, let $g = \nabla F(x)$ and $p = -Hg$. The line search starts with lower bracket $\ell = 0$, upper bracket $h = +\infty$, and trial length $\alpha = 1$. It tests Armijo first and weak Wolfe second:

$$F(x + \alpha p) \le F(x) + c_1 \alpha g^T p, \qquad \nabla F(x + \alpha p)^T p \ge c_2 g^T p. \tag{16}$$

An Armijo rejection sets $h = \alpha$; a Wolfe rejection sets $\ell = \alpha$. The next length is $(\ell + h)/2$ when $h < \infty$, and 2ℓ otherwise. If both tests hold, the length is accepted. This is the unit-first bracketing implementation of Lewis–Overton [1, Algorithm 4.6]. All comparisons in the construction are strict, so strict acceptance tests give the same decisions.

Write the full inverse update as

$$\mathcal{B}(H,g,g^+,\alpha)=\left(I-\frac{sy^T}{y^Ts}\right)H\left(I-\frac{ys^T}{y^Ts}\right)+\frac{ss^T}{y^Ts},\quad s=-\alpha Hg,\quad y=g^+-g. \tag{17}$$

No resetting or rescaling of the stored matrix is performed.

The local counterexample is Theorem 2; we prove it below.

The theorem concerns a specified implementation and a sequence of starting points. It does not assert failure for every nearby start or a positive-measure basin. Its branches have no uniform positive lower curvature bound. The construction differs from the nonconvex prescribed-orbit examples of Mascarenhas [17,18]: here convexity, the unique nearby minimizer, fixed identity initialization and all-query containment are proved simultaneously.

6.2 A finite cycle controlling the entire trial tree

Let (c,d) be the algebraic pair specified in Appendix B. Define

$$g_0=(1,0)^T,\qquad g_1=(-1,-1/8)^T,\qquad g_2=(c,d)^T,\qquad G(z)=\max_{0\le i\le 2} g_i^Tz.$$

The useful coarse bounds are $0.318<c<0.319$ and $-1.026<d<-1.025$. Put

$$\rho=\frac{129(c-8d)}{(16c-127d-16)(127c+16d+129)},\qquad 0.060<\rho<0.061. \tag{18}$$

Set

$$M_0=I,\quad M_1=\begin{pmatrix}129/256 & -1/16\\ -1/16 & 1\end{pmatrix},\qquad M_2=\mathcal{B}(M_1,g_1,g_2,1/4), \tag{19}$$

$$s_0=(-1,0)^T,\quad s_1=(127/1024,1/64)^T,\qquad s_2=\rho(1-c,-d)^T. \tag{20}$$

Finally, let

$$z_0=\frac{1}{1-\rho}\begin{pmatrix}897/1024-\rho(1-c)\\ \rho d-1/64\end{pmatrix},\qquad z_{j+1}=z_j+s_j\quad(j=0,1,2). \tag{21}$$

Lemma 4 (Exact cycle and strict trial decisions) *The matrices in* (19) *are positive definite, and*

$$\mathcal{B}(I,g_0,g_1,1)=M_1,\qquad \mathcal{B}(M_2,g_2,g_0,1/16)=\rho I,\qquad z_3=\rho z_0.$$

At z_j, $j=0,1,2$, the unique active gradient is g_j. With $p_j=-M_jg_j$, the entire line-search query list is Table 2. At every rejection the objective strictly exceeds its current value. At every acceptance,

$$G(z_j)+\tfrac{1}{3}g_j^Ts_j-G(z_{j+1})>0,\qquad g_{j+1}^Ts_j>0>g_j^Ts_j, \tag{22}$$

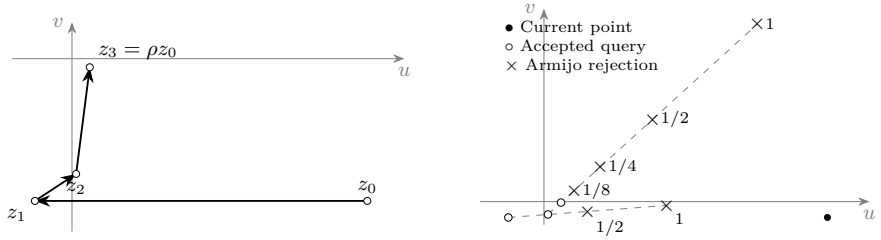
where $g_3=g_0$. All listed queries and the three starting points have a unique active branch and satisfy

$$|g_i^Tz|<3(i=0,1,2),\qquad z_2<2,\qquad G(z)>0. \tag{23}$$

Here z_2 in the middle inequality denotes the second coordinate.

Phase j	Trial α	Active branch	$10^4 G(z_j + \alpha p_j)$	Decision
0	1	1	$(1218, 1219)$	accept
1	1	0	$(3845, 3846)$	reject
1	1/2	0	$(1364, 1365)$	reject
1	1/4	2	$(723, 724)$	accept
2	1	0	$(6687, 6691)$	reject
2	1/2	0	$(3406, 3408)$	reject
2	1/4	0	$(1765, 1767)$	reject
2	1/8	0	$(944, 946)$	reject
2	1/16	0	$(534, 536)$	accept

Table 2 Rational enclosures and the deterministic trial tree for the algebraic seed. Every rejection is an Armijo rejection. The intervals are enclosures, not rounded approximations asserted to be exact values.



(a) Accepted cycle; vertical scale enlarged

(b) All queries; rejected lengths labeled

Fig. 2 The accepted three-step seed cycle (left), with closure $z_3 = \rho z_0$, and every rejected candidate point (right). The rendering uses the rational center of the certified root box. It illustrates the certified trial geometry; it is not a proof of exact equality or of membership in the stable manifold used for the perturbed construction.

The proof of the lemma, including the rational certificate and its soundness, is in Appendix B. It implies the decisions for all parameters in Theorem 2: an accepted decrease is greater than one third of its predicted decrease, and its new directional derivative is positive. No Wolfe rejection is needed, so the query lengths are exactly the successive bisections displayed in the table.

Figure 2 displays this unit-first geometry: phase zero accepts its first query, while the next two phases bisect past the rejected candidates before reaching their accepted points. The right panel includes all those rejected candidates.

Homogeneity and the two cycle closures now give an infinite induction:

$$x_{3m+j} = \rho^m z_j, \qquad H_{3m+j} = \rho^m M_j, \qquad j = 0, 1, 2. \tag{24}$$

Each query in phase j is ρ^m times its base query. Gradients are unchanged, function differences and gradient–step products scale by ρ^m, and (17) scales the updated matrix by ρ^m. Thus every branch choice and every rejection is preserved, as well as acceptance. The iterates converge to zero, but the direction $w = (-1, 16)^T$ satisfies

$$g_0^T w = g_1^T w = -1, \qquad g_2^T w = -c + 16d < -1.$$

Hence $G(tw) = -t$ for $t \geq 0$, so this limit is nonstationary.

Corollary 3 (Coercive polyhedral global failure) *The same deterministic run, with $H_0 = I$, fails for*

$$\overline{G}(u, v) = \max\{u, -u - v/8, cu + dv, v - 17\}.$$

This function is coercive and has unique minimizer $(-1, 16)$, with value -1, while the run converges to the nonstationary origin with value zero.

Proof At every query, (23) and $0 < \rho < 1$ give $v - 17 < 0 < G(u, v)$, so the fourth branch is inactive. The convex combination of branches $0, 1, 3$ with weights $8/17, 8/17, 1/17$ is identically -1. Equality in this lower bound forces all three branches to equal -1, hence $(u, v) = (-1, 16)$; the remaining branch is strictly smaller there. If $\overline{G}(u, v) \le t$, then $u \le t$, $v \le t + 17$ and $-u - v/8 \le t$, which also give lower bounds on both coordinates. Every sublevel is therefore compact.

This corollary concerns global failure at a wrong corner. The next step places such corners arbitrarily near the true minimizer of one objective.

6.3 Scalar interpolation with affine regions at every scale

Let $\theta = 1/16$, $a_n = \theta^n$, $\varepsilon = 1/128$, and $b_* = 2c - 32d \in (33, 34)$. For $b \in \{2, b_*\}$ define

$$t_b = \frac{1 - (1 + \theta + \theta^2)/b}{1 - \theta^2}, \qquad v = \frac{\xi - t_b + \varepsilon}{2\varepsilon}.$$

$$P_b(\xi) = \begin{cases} 0 & \xi \le t_b - \varepsilon, \\ 3v^2 - 2v^3 & |\xi - t_b| < \varepsilon, \\ 1 & \xi \ge t_b + \varepsilon, \end{cases} \qquad J_b(\xi) = \int_0^\xi P_b(t)\, dt.$$

The middle formula for J_b is $2\varepsilon(v^3 - v^4/2)$; its outer formulas are zero and $\xi - t_b$. For $b\theta a_n \le x \le b a_n$, set

$$\varphi_b(x) = (\theta a_n)^3 + b a_n^3 (1 - \theta)\{\theta^2 \xi + (1 - \theta^2) J_b(\xi)\}, \qquad \xi = \frac{x/(b a_n) - \theta}{1 - \theta}. \tag{25}$$

Complete the definition by

$$\varphi_b(x) = \begin{cases} 1 - 3(x + 1) + 3(x + 1)^2 & x < -1, \\ -x^3 & -1 \le x \le 0, \\ 1 + x - b & x \ge b. \end{cases}$$

Lemma 5 (Scalar interpolation) *The functions φ_b are convex, coercive and globally $C^{2,1}$, are nonnegative, and vanish only at zero. They satisfy*

$$0 \le \varphi_b'' \le \max\{6, 102/b\}, \tag{26}$$

$$\varphi_b(b a_n + h) = a_n^3 + a_n^2 h \qquad (|h| \le a_n/2). \tag{27}$$

Their second derivatives are globally Lipschitz with constant

$$L_b = \max\{6, 139264/(5b^2)\}.$$

Proof The ramp is monotone, C^1, and satisfies $0 \le P_b' \le 96$ and $\int_0^1 P_b = 1 - t_b$. On a positive interval,

$$\varphi_b' = a_n^2\{\theta^2 + (1-\theta^2)P_b(\xi)\}, \qquad \varphi_b'' = \frac{a_n(1+\theta)}{b} P_b'(\xi).$$

The identity $\theta^2 + (1-\theta^2)(1-t_b) = (1+\theta+\theta^2)/b$ gives matching values $\varphi_b(ba_n) = a_n^3$. The derivative is constant near both ends of each interval, so the first two derivatives also match, with values a_n^2 and zero at ba_n.

As $x \downarrow 0$, the bounds $a_n \le x/(b\theta)$ on its interval give $\varphi_b(x) = O(x^3)$, $\varphi_b'(x) = O(x^2)$ and $\varphi_b''(x) = O(x)$. These match the negative cubic at zero. The formulas also match through second order at -1 and b. The derivative is negative on the negative half-line and positive on the positive half-line; the outer formulas diverge to infinity. This proves convexity, coercivity and the minimum assertions.

The constant-derivative lengths below and above ba_n, divided by a_n, are

$$b(1-\theta)(1-t_b-\varepsilon), \qquad \frac{b(1-\theta)(t_b-\varepsilon)}{\theta}.$$

Both exceed $1/2$ for $2 \le b \le 34$, by substitution in the displayed rational formulas; the affine outer extension covers the upper side when $n = 0$. This proves (27). Finally $|P_b''| \le 3/(2\varepsilon^2)$ in each ramp, so differentiation once more gives $|\varphi_b'''| \le 139264/(5b^2)$ there. The negative cubic has third derivative of magnitude six. Continuity of φ_b'' permits summing the Lipschitz estimates over finitely many joins away from zero; taking limits at zero covers the infinitely many accumulating joins and gives the global bound.

6.4 One objective and a uniform induction over all queries

Define

$$F(u,v) = \max\{\varphi_2(u), \varphi_2(-u-v/8), \varphi_{b_*}(cu+dv), v\}. \tag{28}$$

The Hessian of each of the first three branches is a nonnegative scalar multiple of $g_i g_i^T$. Equation (26) gives the upper bounds 51, $51(65/64) < 52$, and 12, respectively, on their operator norms; the last branch has zero Hessian. Their Hessians are globally Lipschitz because φ_b'' is Lipschitz and the arguments are linear. The first two scalar arguments are independent, so their coercivity makes every sublevel of F compact. Nonnegativity and their unique scalar zeros show $F(x) = 0$ exactly at $x = 0$. At zero the first three gradients vanish and the last is $(0,1)^T$, giving $F'(0;h) = \max\{0, h_2\}$; thus the minimizer is nonsmooth.

Let $p_n = a_n(2,-32)^T$. The first two scalar arguments at p_n equal $2a_n$, and the third equals $b_* a_n$. Consequently

$$F(p_n + h) = a_n^3 + a_n^2 G(h) \quad \text{if } |g_i^T h| \le a_n/2 (i = 0,1,2). \tag{29}$$

Indeed, the first three branches are on their affine regions and each has value at least $a_n^3/2$. The first two slab inequalities imply $|h_1| \le a_n/2$ and $|h_2| \le 8a_n$, so

the fourth branch is negative. In particular $F(p_n) = a_n^3$ and $F'(p_n; (-1, 16)^T) = -a_n^2 < 0$.

Proof (Proof of Theorem 2) For $n \geq 1$ take $x_0^{(n)} = p_n + a_n^2 z_0$ and $H_0 = I$. We claim that

$$x_{3m+j}^{(n)} = p_n + a_n^2 \rho^m z_j, \qquad H_{3m+j}^{(n)} = \rho^m M_j, \qquad j = 0, 1, 2. \tag{30}$$

At a current point of this form its gradient is $a_n^2 g_j$. Every trial on the finite candidate list therefore has the form $p_n + a_n^2 \rho^m z_{\text{trial}}$. Equation (23) gives

$$|g_i^T (a_n^2 \rho^m z_{\text{trial}})| < 3a_n^2 \leq 3a_n/16 < a_n/2.$$

Thus (29) is valid at every candidate query, not only the final accepted one. Unique activity is preserved, and

$$F(p_n + a_n^2 \rho^m z_{\text{trial}}) = a_n^3 + a_n^4 \rho^m G(z_{\text{trial}}), \qquad \nabla F = a_n^2 g_{\text{active}}.$$

The additive constant cancels in Armijo. Function differences and gradient–step products are the seed quantities multiplied by the same positive factor $a_n^4 \rho^m$. The actual search therefore traverses exactly the candidate list and makes the same strict decisions. In (17), the projection factors are unchanged and the rank-one term has factor ρ^m. The cycle closures prove the next point and matrix identities, completing the induction.

The finite matrices are positive definite. Equation (30) gives convergence to p_n and strict descent to $a_n^3 > 0$. Since the base query set is finite, the norm of every query is bounded by

$$a_n \|(2, -32)\| + a_n^2 \max_{\text{base points and queries}} \|z\|,$$

which tends to zero. The starting points also tend to zero. All other claims were proved directly from (28).

7 Verification and the remaining boundary

The formal record uses Lean 4.19.0 with pinned mathlib. A fresh incremental project build and declaration-level audit were run on 12 September 2026. The audited main and supplemental correction modules contain 9,781 safe declarations, including 7,954 theorems. Their type, proof-value and structural dependencies were inspected transitively. Only propositional extensionality (`propext`), classical choice (`Classical.choice`) and quotient soundness (`Quot.sound`) are permitted axiom dependencies. No `sorryAx`, custom mathematical axiom, unchecked external calculation, or reachable unsafe or partial implementation artifact passed the audit. These statements describe an actual run, not a restored historical count.

The exact theorem contracts determine the formal scope. For example, the principal-minor result concerns actual matrices and determinants, and

the scalar hierarchy is derived from it. Conditional sequence lemmas alone would not establish that bridge. Similarly, the negative exports include the implemented search decisions and the induction over all queries; a finite numerical trajectory would not suffice. The companion supplies the full export lists, source and theorem correspondence, pinned reproduction commands, and the audit logs. The article's rational certificate and mathematical proofs are readable independently of Lean.

Formal checking does not prove consistency of the foundational axioms, novelty, fidelity of a natural-language paraphrase, or floating-point robustness. The standard kernel and execution environment remain trusted. The correspondence and the three editorial review passes are author-side reviews, not independent human peer review. Stable-manifold membership in Appendix C is an existence theorem and is not inferred from rounded forward shooting.

The results leave a focused boundary. Local finite polyhedral geometry with smooth extensions controls inward model error, while smooth convex branch formulas alone can encode bad configurations accumulating at the minimizer. Uniform lower curvature does not repair the separate $C^{1,1}$ construction. The present results do not decide the local problem for finite uniformly strongly convex C^2 branches through the optimizer, or classify all convex nonsmooth objectives by affine-critical or accessible-jump geometry.

Reproducibility and AI assistance. The accompanying source tree contains the article, complete companion, figure-generation code, exact certificates and Lean project. Large language models assisted theorem development, proof reconstruction, formalization, literature searches, software, and writing. This assistance was substantive; it was not limited to grammar editing. Compiler and arithmetic checks provide the specific evidence described above. Human authors must review the claims, citations, novelty assessment and disclosure before submission. This author-review edition asserts no author consent, independent human review, affiliations, grants, or public archival identifier.

A The BFGS compression calculation

Lemma 6 *Under the assumptions of Theorem 4, let Z have j orthonormal columns, $1 \le j \le n$. Then*

$$\det(Z^T H_+ Z) \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q} + \frac{\alpha}{\beta} \det(Z^T H Z).$$

For $j = n$, $\det H_+ = (\alpha/\beta) \det H$.

Proof Put $v = Hg$, $\kappa = g^T H y = -\beta q$ and $P = I - vy^T/\kappa$. Since $y^T s = \alpha\beta q$, the update becomes

$$H_+ = \mathcal{F} + \frac{\alpha}{\beta q} v v^T, \qquad \mathcal{F} = PHP^T.$$

Write

$$A = Z^T H Z \succ 0, \quad a = Z^T H g, \quad b = Z^T H y, \quad r = y^T H y,$$
$$u = a^T A^{-1} a, \quad w = a^T A^{-1} b, \quad z = b^T A^{-1} b.$$

Then

$$Z^T \mathcal{F} Z = A - [a\ b] \begin{pmatrix} -r/\kappa^2 & 1/\kappa \\ 1/\kappa & 0 \end{pmatrix} [a\ b]^T.$$

The identity $\det(I + UV) = \det(I + VU)$ reduces its determinant to a 2×2 determinant. Expanding gives

$$\det(Z^T \mathcal{F} Z) = \frac{\det A}{\kappa^2}\big((\kappa - w)^2 + u(r - z)\big). \tag{31}$$

Schur complements give, independently,

$$\det \mathrm{Gram}_H(y, Z) = \det A(r - z),$$
$$\det \mathrm{Gram}_H(g, y, Z) = \det A\big((q - u)(r - z) - (\kappa - w)^2\big).$$

Both are Gram determinants and hence nonnegative, including when their column families are dependent. Combining these identities yields

$$\det(Z^T \mathcal{F} Z) = \frac{q \det \mathrm{Gram}_H(y, Z) - \det \mathrm{Gram}_H(g, y, Z)}{\beta^2 q^2} \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q}.$$

Adding $\rho a a^T$ to the compression changes the coefficient r/κ^2 by ρ. The same 2×2 expansion therefore changes its determinant by exactly $\rho \det A\, u$. No invertibility of $Z^T \mathcal{F} Z$ is used. For $\rho = \alpha/(\beta q)$, the Schur complement of A in $\mathrm{Gram}_H(g, Z)$ gives $0 \le u \le q$, so the increase is at most $(\alpha/\beta) \det A$. This proves the inequality.

When $j = n$, Z is an orthogonal basis and $u = q$, $w = \kappa$, $z = r$. Equation (31) then vanishes, while the rank-one increase is exactly $(\alpha/\beta) \det A$. Orthogonal invariance gives the stated determinant identity.

B The algebraic seed and its finite rational certificate

This appendix gives the objects and finite inequalities needed for Lemma 4. All decimals defining a box are terminating rationals. No rounded point is substituted for a root or an infinite orbit.

B.1 Existence and uniqueness of the defining root

Put

$$U = 127c + 16d, \qquad V = 16c - 127d - 16, \qquad R = c - 8d,$$
$$K = 8c^2 + 8d^2 + d - 8, \qquad T = c^2 - c + d^2, \qquad Y = (c-1)^2 + d^2,$$

and let

$$P(c,d) = \begin{pmatrix} 264192RK - UV(U+129) \\ 16VTK - RUY \end{pmatrix}. \tag{32}$$

The root is specified by $P(c,d) = 0$ and $\|(c,d) - \bar{r}\|_\infty \le 10^{-30}$, where

$$\bar{r}_1 = 0.31848142762711863064645131250290703717912205023605,$$
$$\bar{r}_2 = -1.02523211967574872548135087001300414284511909318444.$$

Lemma 7 (Rational root certificate) *This closed box contains exactly one root. On the box all of $U, V, R, K, T, Y, U + 129$ are positive, and* (18) *holds.*

Proof Let $D = DP(\bar{r})$ and define its inverse by the ordinary two-by-two adjugate formula $A = \operatorname{adj}(D)/\det D$. The following finite rational certificate uses only the coefficients of (32) and the two displayed rationals:

$$\det D \neq 0, \qquad \max_i |(AP(\bar{r}))_i| < 10^{-50}, \qquad \max_i \sum_{2 \le |\nu| \le 5} |a_{i,\nu}|\,|\nu| \le 7000, \tag{33}$$

where, for $\nu = (\nu_1, \nu_2)$,

$$a_{i,\nu} = -\frac{1}{\nu_1!\nu_2!} \sum_{j=1}^{2} A_{ij}\, \partial^\nu P_j(\bar{r}).$$

These formulas specify every rational in the certificate: the index set contains the eighteen monomials of degrees two through five. The bounds in (33) follow by substituting the rational center, differentiating the displayed polynomials, and clearing positive denominators in the three comparisons. Thus they are finite integer inequalities, rather than tolerances for a numerically computed root.

For completeness, their analytic consequence follows directly. The centered residual map $N(h) = h - AP(\bar{r} + h)$ has the exact expansion

$$N_i(h) = -(AP(\bar{r}))_i + \sum_{2 \le |\nu| \le 5} a_{i,\nu} h^\nu;$$

its linear part vanishes because $AD = I$. On $\|h\|_\infty, \|k\|_\infty \le r = 10^{-30}$, telescoping the factors of each monomial gives $|h^\nu - k^\nu| \le |\nu| r \|h - k\|_\infty$, since $r \le 1$ and $|\nu| \ge 2$. Hence N has Lipschitz constant at most $7000r$ and

$$\|N(h)\|_\infty \le 10^{-50} + 7000 r^2 < r.$$

The closed box is complete, so contraction gives a unique fixed point. Since A is invertible, this is exactly a root of P. This is the standard preconditioned-residual principle; compare Rump [19, Theorem 13.3].

The root box lies in $318/1000 < c < 319/1000$ and $-1026/1000 < d < -1025/1000$. Substitution gives positivity of the stated factors. For example,

$$\frac{2397}{100} \le U \le \frac{24113}{1000}, \quad \frac{119263}{1000} \le V \le \frac{59703}{500}, \quad \frac{4259}{500} \le R \le \frac{8527}{1000}.$$

Multiplying positive endpoints in $129R/[V(U+129)]$ gives $60/1000 < \rho < 61/1000$. In particular all divisions used below are valid.

B.2 The matrix identity

Proof (Proof of the algebraic part of Lemma 4) The first update in (19) is immediate from $s_0 = (-1, 0)^T$ and $g_1 - g_0 = (-2, -1/8)^T$. For arbitrary (c, d) with the positive factors above, put

$$\beta = \frac{66048RK}{UV(U+129)}, \quad \gamma = \frac{VTK}{RUY}, \quad \widetilde{\rho} = \frac{2064TK}{U(U+129)Y}.$$

The rational identities obtained from (17) are

$$\mathcal{B}(\mathcal{B}(M_1, g_1, g_2, \beta), g_2, g_0, \gamma) = \widetilde{\rho} I, \qquad -\gamma \mathcal{B}(M_1, g_1, g_2, \beta) g_2 = \widetilde{\rho}(g_0 - g_2). \tag{34}$$

Here is a way to check the identities without guessing matrix entries. The middle update is $C + \beta E$, because its two projection factors are independent of the step length and its rank-one term is linear in that length. Requiring $(C+\beta E)g_2$ to be parallel to $g_0 - g_2$ gives a linear equation for β; substitution of M_1, g_1, g_2 gives the displayed value. For a unit vector $w \perp (g_0 - g_2)$, substitution gives $w^T(C + \beta E)w = \widetilde{\rho}$. With the displayed γ, the last step is $\widetilde{\rho}(g_0 - g_2)$. Its projection factors are the orthogonal projector onto $(g_0 - g_2)^{\perp}$, and its rank-one term is $\widetilde{\rho}$ times the complementary projector. Their sum is $\widetilde{\rho} I$, proving (34). Each substitution is a rational identity and can equivalently be checked by clearing their nonzero denominators and expanding.

Equations (32) give exactly $\beta = 1/4$ and $\gamma = 1/16$. Moreover,

$$\widetilde{\rho} - \rho = \frac{2064R}{V(U+129)} \left(\gamma - \frac{1}{16} \right) = 0.$$

This gives the matrix and displacement closures. The formula for z_0 gives $s_0 + s_1 + s_2 = -(1-\rho)z_0$. The strict inequalities below imply $(g_{j+1} - g_j)^T s_j > 0$ at each phase. Positive definiteness follows inductively from the full update: its two nonnegative quadratic terms cannot both vanish on a nonzero vector when $y^T s > 0$.

B.3 Sound enclosure of every query

An interval $[a, b]$ encloses the corresponding real quantity. Addition, negation, multiplication and reciprocal are evaluated by

$$[a,b]+[c,d]=[a+c,b+d], \qquad -[a,b]=[-b,-a],$$
$$[a,b][c,d]=[\min(ac,ad,bc,bd),\max(ac,ad,bc,bd)],$$

and $[a, b]^{-1} = [1/b, 1/a]$ when $0 \notin [a, b]$. These inclusion rules follow from order on the reals; induction on a rational expression proves the soundness of every resulting bound. All endpoints used here are rational, so no directed floating-point rounding is needed.

For the query calculation it suffices to use the larger box

$$c \in \frac{[318481, 318482]}{10^6}, \quad d \in \frac{[-1025233, -1025232]}{10^6}, \quad \rho \in \frac{[6019, 6021]}{10^5}. \tag{35}$$

The first two inclusions follow from the root box; substitution in (18), with positive denominator, gives the third. Evaluate z_j from (21) and use

$$p_0 = (-1, 0)^T, \quad p_1 = (127/256, 1/16)^T, \quad p_2 = 16\rho(1 - c, -d)^T.$$

For a row with active index i, set $q = z_j + \alpha p_j$ and evaluate the following rational expressions by the inclusion rules:

$$(g_i - g_k)^T q (k \neq i), \quad -\alpha g_j^T p_j, \quad \begin{cases} g_j^T z_j + \alpha g_j^T p_j/3 - g_i^T q, & \text{acceptance,} \\ g_i^T q - g_j^T z_j, & \text{rejection.} \end{cases}$$

For an acceptance also evaluate $\alpha g_i^T p_j$. Every expression in its row is strictly greater than the following positive rational lower bound, in the order of Table 2:

$$\frac{1}{10^4}(729, 2417, 145, 75, 5964, 2682, 1041, 221, 20). \tag{36}$$

The same interval operations give the value enclosures in that table. For the three base points, the designated active gaps are positive as well. Evaluating $3 \pm g_i^T q$ for every i, $2 - q_2$, and the designated active value $g_i^T q$ gives positive lower bounds for all base points and all nine queries. This proves (23). All inputs, expressions, interval rules and output bounds of this finite certificate are now specified. The strict active gaps also give differentiability on a neighborhood of each query. Together with the matrix identities, these calculations complete the proof of Lemma 4.

C Positive lower curvature through a perturbed cycle

Theorem 3 uses a different regularity balance from Theorem 2. Adding a positive quadratic changes the seed orbit, so exact homogeneity alone no longer supplies a trajectory. We first prove persistence on a stable manifold and then copy one entire perturbed orbit at all scales. Throughout this appendix the line-search constants are $c_1 = 10^{-4}$ and $c_2 = 1/2$.

C.1 A strongly convex global example

Use the fixed slopes of Appendix B and let

$$Q_\mu(x) = \tfrac{\mu}{2}\|x\|^2 + G(x), \qquad \mu > 0.$$

The vector

$$\bar{g} = \frac{(1,-16)^T}{257} = \frac{129}{257}g_0 + \frac{128}{257}g_1$$

is the least-norm point of $\operatorname{conv}\{g_0, g_1, g_2\}$: $g_0^T\bar{g} = g_1^T\bar{g} = \|\bar{g}\|^2 = 1/257$, whereas $g_2^T\bar{g} > 1/257$. Therefore the unique minimizer is

$$x_* = -\bar{g}/\mu, \qquad \min Q_\mu = -1/(514\mu). \tag{37}$$

At x_* the first two branches tie with distinct gradients, so the minimizer is nonsmooth. At zero the subdifferential is the nonstationary triangle of the original seed.

Proposition 1 (A perturbed bad orbit) *For every $\mu > 0$ and every neighborhood V of zero, some initial state (x_0, H_0), $H_0 \succ 0$, produces a deterministic full-BFGS run on Q_μ converging to zero, with positive objective values decreasing to zero. Its accepted lengths repeat $1, 1/4, 1/16$, and all accepted and rejected query points are differentiable and belong to V.*

To prove this proposition, use normalized coordinates

$$x = \tau z, \qquad H = \tau M(b,e), \qquad M(b,e) = \begin{pmatrix} 1 & b \\ b & e \end{pmatrix}, \qquad \tau > 0.$$

The boundary seed is $P_* = (z_0, 0, 1, 0)$ in coordinates (z, b, e, τ). Keep the input τ fixed during three substeps. Starting with $(Z, M) = (z, M(b,e))$, for $j = 0, 1, 2$ perform

$$\begin{aligned} &\gamma_j = g_j + \mu\tau Z, \qquad S = -\alpha_j M\gamma_j, \qquad Z^+ = Z + S, \\ &Y = g_{j+1} - g_j + \mu\tau S, \\ &M^+ = \left(I - \frac{SY^T}{Y^TS}\right) M \left(I - \frac{YS^T}{Y^TS}\right) + \frac{SS^T}{Y^TS}. \end{aligned} \tag{38}$$

with cyclic gradient indices and $\alpha = (1, 1/4, 1/16)$. For the resulting (Z_3, M_3), let $r = (M_3)_{11}$ and define

$$\mathcal{T}_\mu(z, b, e, \tau) = (Z_3/r, (M_3)_{12}/r, (M_3)_{22}/r, \tau r)\,. \tag{39}$$

The seed closure gives $\mathcal{T}_\mu(P_*) = P_*$ and $r(P_*) = \rho$. All denominators are nonzero there. Hence the map is real analytic on an open neighborhood of P_*, using negative τ only to define this analytic extension.

Lemma 8 (Implementation and hyperbolicity) *On a sufficiently small neighborhood of P_*, positive-τ states follow exactly the three query lists of Table 2 under the actual algorithm, and* (39) *is their return map. Its derivative has eigenvalues*

$$\rho^{-1}, \quad \rho^{-1}, \quad \lambda_s, \quad \lambda_u, \quad \rho, \qquad .038 < \lambda_s < .039, \quad 1.564 < \lambda_u < 1.565.$$

The stable eigenspace contains a vector whose τ coordinate is one.

Proof At a query τZ, the value divided by τ is $G(Z) + \mu\tau\|Z\|^2/2$ and the gradient is $g_i + \mu\tau Z$. The common quadratic does not affect which slope is active. All finite branch gaps and normalized search comparisons are continuous in the normalized state. The strict seed margins therefore persist on a common neighborhood. The physical step and secant are exactly the ones in (38) after restoring the scale, proving agreement with the actual update and the entire search tree.

At $\tau = 0$ the matrix return is independent of z. Let J be the derivative at $(b, e) = (0, 1)$ of its normalized shape map. The following rational enclosures specify the spectral certificate:

$$10^{10} J \ \in \begin{pmatrix} [11123986249, 11123986253] & [6459701025, 6459701029] \\ [7515015701, 7515015708] & [4905136882, 4905136890] \end{pmatrix}. \tag{40}$$

They are obtained by differentiating the rational substeps (38) at $\tau = 0$, then applying the interval rules of Appendix B to

$$c \in [318481427627118, 318481427627119]/10^{15},$$
$$d \in [-1025232119675749, -1025232119675748]/10^{15}.$$

Both intervals contain the defining root box. This is a finite rational derivative calculation: for a product use $D(ab) = a\,Db + b\,Da$, and for an inverse use $D(a^{-1}) = -a^{-2}Da$, with denominators bounded away from zero. Substitution of (40) in $p(\lambda) = \lambda^2 - \operatorname{tr}(J)\lambda + \det J$ gives

$$p(.038) > 0, \qquad p(.039) < 0, \qquad p(1.564) < 0, \qquad p(1.565) > 0.$$

These rational sign comparisons locate two distinct real roots and exhaust the degree-two polynomial; no numerical eigenvector is used.

At $\tau = 0$, $Z_3 = z + v(b, e)$ and r is independent of z. The last coordinate of the return is τr. Thus its full derivative has the block form

$$D\mathcal{T}_\mu(P_*) = \begin{pmatrix} \rho^{-1} I_2 & C & \mu a \\ 0 & J & \mu b' \\ 0 & 0 & \rho \end{pmatrix}.$$

The unspecified blocks are finite derivatives of the displayed rational map and do not affect its spectrum. None of the five eigenvalues is zero or has modulus one. Moreover ρ is not an eigenvalue of the upper left four-dimensional block. Assigning last coordinate one and solving the remaining four linear equations gives a ρ-eigenvector. This proves the last assertion and the lemma.

Proof (Proof of Proposition 1) The inverse function theorem makes the return map a local diffeomorphism. The classical local stable-manifold theorem provides a two-dimensional smooth stable manifold through P_*, tangent to the stable eigenspace; (Dyatlov, *Notes on hyperbolic dynamics*, 2018 preprint, arXiv:1805.11660v1) gives an exposition of this principle and its graph transform proof. By the last assertion of Lemma 8, this manifold contains points with arbitrarily small positive τ. Choose such a point whose forward return orbit stays in the implementation neighborhood and tends to P_*. Shrinking the neighborhood keeps $r > 0$ and $M(b, e) \succ 0$, so $\tau^+ = \tau r$ preserves positivity. The actual intermediate matrices remain positive definite by positive curvature.

The physical iterates are $x_{3m} = \tau_m z_m \to 0$. Intermediate points and all queries are τ_m times bounded continuous functions of the normalized state, so they tend to zero too. Their normalized objective values stay positive by the seed's strict positivity. Armijo makes the values decrease to zero, which exceeds (37). The initial stable-manifold point can be taken sufficiently close to P_* that the entire local stable orbit, and hence every physical query, lies in the prescribed V. This proves the proposition. The initial matrix is supplied by the stable-manifold point, not presumed to be identity or obtained by rounded forward shooting.

C.2 An accumulation embedding with four strongly convex branches

Fix

$$\mu = 1/4096, \qquad c_* = (2, -32)^T, \qquad q_i = g_i - \mu c_*,$$
$$b_i = q_i^T c_*, \qquad r = 1/128, \qquad a_n = r^n.$$

Here $b_0 = b_1 = 1791/1024$ and $1.7 < b_i < 34$. For each of the two distinct values of b_i, set

$$t_b = \frac{1 - (1 + r)/b}{1 - r}, \qquad \varepsilon = 1/1024.$$

Use the same cubic ramp P_b and primitive J_b as in (25), with this center and width. For $ba_{n+1} \le x \le ba_n$ put

$$\psi_b(x) = (ra_n)^2 + ba_n^2(1 - r)\{r\xi + (1 - r)J_b(\xi)\}, \qquad \xi = \frac{x/(ba_n) - r}{1 - r}. \quad (41)$$

Set $\psi_b(0) = 0$, $\psi_b(x) = x^2/2$ for $x < 0$, and $\psi_b(x) = 1 + x - b$ for $x \ge b$.

On each positive interval,

$$\psi_b' = a_n\{r + (1 - r)P_b(\xi)\}, \qquad \psi_b'' = P_b'(\xi)/b, \qquad 0 \le P_b' \le 768.$$

The identity $r + (1 - r)(1 - t_b) = (1 + r)/b$ matches node values and derivatives. Their values are a_n^2 and a_n, respectively. The positive derivative is monotone and constant near every node. Its derivative slopes are uniformly bounded, so summing over intervals and taking a limit at zero proves that ψ_b is globally

$C^{1,1}$, convex and coercive, nonnegative with unique zero at the origin. The two constant-derivative lengths, divided by a_n, are

$$b(1-r)(1-t_b-\varepsilon), \qquad b(1-r)(t_b-\varepsilon)/r,$$

both greater than $1/2$ for these values of b. Consequently

$$\psi_b(ba_n+h) = a_n^2 + a_n h \qquad (|h| \le a_n/2). \tag{42}$$

At zero, the difference quotients of ψ_b' along ba_n and $(b+1/4)a_n$ equal $1/b$ and $1/(b+1/4)$, respectively. Thus its second derivative at zero does not exist.

Define

$$\begin{aligned} f_i(x) &= \tfrac{\mu}{2}\|x\|^2 + \psi_{b_i}(q_i^T x) \qquad (i=0,1,2),\\ f_3(x) &= \tfrac{\mu}{2}\|x\|^2 + x_2, \qquad F = \max_i f_i. \end{aligned} \tag{43}$$

Each branch is globally μ-strongly convex and $C^{1,1}$. For the first three branches, the gradient-Lipschitz constants are at most $L_i = \mu + \max\{1, 768/b_i\}\|q_i\|^2$. Direct substitution of the rational root bounds gives $L_0 < 439$, $L_1 < 446$ and $L_2 < 27$; hence 512 is a common constant, also covering the fourth branch. Since $F(x) \ge \mu\|x\|^2/2$ with equality zero at the origin, the sublevels are compact and the minimizer is unique. The first three gradients at zero vanish, whereas the fourth is $(0,1)^T$, so the minimizer is nonsmooth.

At a differentiable point of a finite maximum of L-smooth functions, all active gradients agree. Inactive branches stay inactive locally; taking the maximum of the active upper Taylor bounds therefore gives the local upper Taylor bound with constant L. Applying this with $L = 512$ and combining it with strong convexity proves

$$\mu I \preceq \nabla^2 F(x) \preceq 512 I \quad \text{whenever } F \text{ is twice differentiable.}$$

This does not require a globally Lipschitz gradient for the maximum itself.

C.3 Copying one entire orbit and fixing the initial matrix

Let $p_n = a_n c_*$ and $C_* = 1 + \mu\|c_*\|^2/2$. For y in a fixed sufficiently small neighborhood of zero, (42) yields

$$F(p_n + a_n y) = a_n^2\{C_* + Q_\mu(y)\} \qquad (n \ge 1). \tag{44}$$

Indeed, $|q_i^T y| \le 1/2$ puts each old branch in its affine scalar region, and expansion of the common quadratic uses $q_i + \mu c_* = g_i$. The fourth branch is inactive there: its residual $a_n(-32 + y_2)$ is negative, while each old scalar residual is at least $a_n^2/2$.

Choose, by Proposition 1, one bad orbit $(\bar{x}_k, \bar{H}_k)$ all of whose queries lie in this neighborhood. For every n, start at $p_n + a_n\bar{x}_0$ with the same matrix $\bar{H}_0$. Induction on the actual query tree gives

$$x_k^{(n)} = p_n + a_n\bar{x}_k, \qquad H_k^{(n)} = \bar{H}_k.$$

Function differences and directional products scale by a_n^2; gradients and steps scale by a_n. Both scale factors cancel in the inverse update, so each rejection, acceptance and updated matrix is preserved. Equation (44) applies to every query because the whole chosen seed orbit lies in its domain. Thus $x_k^{(n)} \to p_n$ and $F(x_k^{(n)}) \downarrow a_n^2 C_* > 0$. The three active gradients at p_n are $a_n g_i$, whose convex hull excludes zero. All query norms are at most $a_n(\|c_*\| + \sup_{\text{seed queries}} \|\bar{x}\|) \to 0$.

Finally set $A = \bar{H}_0^{1/2}$ and $\widetilde{F}(z) = F(Az)$. The coordinate identities

$$\widetilde{g} = A^T g, \qquad \widetilde{H} = A^{-1} H A^{-T}, \qquad \widetilde{s} = A^{-1} s$$

preserve both scalar line-search tests and (17); see also [1, Section 2]. They give $\widetilde{H}_0 = I$ for this one fixed transformation. Strong-convexity and smoothness constants become $\underline{\mu} = \mu\lambda_{\min}(\bar{H}_0)$ and $\overline{L} = 512\lambda_{\max}(\bar{H}_0)$. All locality and differentiability statements persist under the invertible linear map. This proves the finite-branch assertion in Theorem 3.

C.4 The constant classical-Hessian alternative

Replace the ramp in (41) by $P_b(\xi) = \mathbf{1}_{\{\xi > t_b\}}$, its primitive by $(\xi - t_b)_+$, and the negative scalar extension by $-x$. The resulting scalar function is convex, countably piecewise affine, and retains the node values, node slopes and affine regions (42). Hence the same orbit-copy proof applies. The new objective is $F(x) = \mu\|x\|^2/2 + p(x)$, with convex countably piecewise-affine p.

At any nonzero point, a scalar branch with $q_i^T x = 0$ is strictly inactive: its residual is zero, whereas the independence of q_0, q_1 ensures some scalar residual is positive. The possibly active scalar arguments are nonzero and meet only finitely many breakpoints in a small neighborhood. Thus p is locally finite-polyhedral away from zero. At a differentiable point it is locally affine. At zero the directional derivative along c_* is zero, since all $b_i > 0$ and the positive scalar parts are $O(t^2)$; along $-c_*$ it is positive, since the negative scalar parts are linear. Thus zero is nonsmooth. Therefore $\nabla^2 F = \mu I$ wherever the classical Hessian exists. After the same fixed coordinate transformation this Hessian is $\mu\bar{H}_0 \succ 0$. This proves the remaining alternative of Theorem 3 without asserting four smooth branches for that alternative.

References

1. A. S. Lewis and M. L. Overton. *Nonsmooth optimization via quasi-Newton methods.* Mathematical Programming 141 (2013), 135–163; published online 16 February 2012. https://doi.org/10.1007/s10107-012-0514-2.
2. D. Wachsmuth and G. Wachsmuth. *A simple proof of the Baillon–Haddad theorem on open subsets of Hilbert spaces.* Journal of Convex Analysis 30(4) (2023), 1319–1328. https://journalofconvexanalysis.com/articles/jca30059/.
3. B. Gebken. *Technical Results on the Convergence of Quasi-Newton Methods for Nonsmooth Optimization.* Journal of Optimization Theory and Applications 209, article 57 (8 May 2026). https://doi.org/10.1007/s10957-026-03008-z.

4. J. Bolte, A. Daniilidis, A. S. Lewis and M. Shiota. *Clarke Subgradients of Stratifiable Functions*. SIAM Journal on Optimization 18(2) (2007), 556–572. https://doi.org/10.1137/060670080.
5. A. S. Lewis, J. Liang and T. Tian. *Partial Smoothness and Constant Rank*. SIAM Journal on Optimization 32(1) (2022), 276–291; published online 17 March 2022. https://doi.org/10.1137/19M1237909.
6. A. Banerjee, S. Merugu, I. S. Dhillon and J. Ghosh. *Clustering with Bregman Divergences*. Journal of Machine Learning Research 6 (2005), 1705–1749. https://www.jmlr.org/papers/v6/banerjee05b.html.
7. J. Guo and A. S. Lewis. *Nonsmooth Variants of Powell's BFGS Convergence Theorem*. SIAM Journal on Optimization 28(2) (2018), 1301–1311. https://doi.org/10.1137/17M1121883.
8. F. Schöpfer. *Linear Convergence of Descent Methods for the Unconstrained Minimization of Restricted Strongly Convex Functions*. SIAM Journal on Optimization 26(3) (2016), 1883–1911. https://doi.org/10.1137/140992990.
9. J. Nocedal. *Theory of algorithms for unconstrained optimization*. Acta Numerica 1 (1992), 199–242. https://doi.org/10.1017/S0962492900002270.
10. R. H. Byrd and J. Nocedal. *A Tool for the Analysis of Quasi-Newton Methods with Application to Unconstrained Minimization*. SIAM Journal on Numerical Analysis 26(3) (1989), 727–739. https://doi.org/10.1137/0726042.
11. M. J. D. Powell. *Some global convergence properties of a variable metric algorithm for minimization without exact line searches*.
In R. W. Cottle and C. E. Lemke (eds.), *Nonlinear Programming*. SIAM–AMS Proceedings 9, American Mathematical Society (1976), 53–72.
12. Q. Jin, R. Jiang and A. Mokhtari. *Non-asymptotic Global Convergence Analysis of BFGS with the Armijo-Wolfe Line Search*. Advances in Neural Information Processing Systems 37 (NeurIPS 2024). https://doi.org/10.52202/079017-0536.
13. P. L. Combettes and B. C. Vũ. *Variable metric quasi-Fejér monotonicity*. Nonlinear Analysis: Theory, Methods & Applications 78 (2013), 17–31. https://doi.org/10.1016/j.na.2012.09.008.
14. A. S. Lewis. *Active sets, nonsmoothness, and sensitivity*. SIAM Journal on Optimization 13(3) (2002), 702–725. https://doi.org/10.1137/S1052623401387623.
15. A. S. Lewis and S. Zhang. *Nonsmoothness and a Variable Metric Method*. Journal of Optimization Theory and Applications 165 (2015), 151–171; published online 7 August 2014. https://doi.org/10.1007/s10957-014-0622-7.
16. A. Asl and M. L. Overton. *Analysis of limited-memory BFGS on a class of nonsmooth convex functions*. IMA Journal of Numerical Analysis 41(1) (2021), 1–27; advance access 18 January 2020. https://doi.org/10.1093/imanum/drz052.
17. W. F. Mascarenhas. *The BFGS method with exact line searches fails for non-convex objective functions*. Mathematical Programming 99 (2004), 49–61; published online 7 May 2003. https://doi.org/10.1007/s10107-003-0421-7.
18. W. F. Mascarenhas. *On the divergence of line search methods*. Computational and Applied Mathematics 26(1) (2007), 129–169. https://www.scielo.br/j/cam/a/JBFzLzr4PNdcqKHJRRS4mHf/.
19. S. M. Rump. *Verification methods: Rigorous results using floating-point arithmetic*. Acta Numerica 19 (2010), 287–449. https://doi.org/10.1017/S096249291000005X.
20. R. T. Rockafellar. *Convex Analysis*. Princeton Mathematical Series 28. Princeton University Press, Princeton, NJ (1970). https://doi.org/10.1515/9781400873173.